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OF
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Q1: Data Acquisition

Historical daily price data were obtained using the `yfinance` Python library, which provides a programmatic interface to Yahoo Finance market data. We downloaded adjusted closing prices for seven large, liquid stocks—Apple (AAPL), Microsoft (MSFT), NVIDIA (NVDA), Amazon.com (AMZN), JPMorgan Chase (JPM), Barclays (BARC.L) and Johnson & Johnson (JNJ)—and for the S&P 500 index (^GSPC) over the period from 1 September 2024 to 1 September 2025. The data were requested at a daily frequency using `auto_adjust=True` so that prices are adjusted for corporate actions and are suitable for return calculations. The downloaded series were combined into a single **DataFrame** with trading dates as the index and tickers as columns. To verify that the dataset is consistent across all assets, we examined the date range, counted missing observations for each ticker and compared the number of rows before and after dropping any rows containing missing values. Because the universe includes both US- and UK-listed equities, some differences in exchange holidays led to a small number of missing prices; to ensure reliability in later analysis, we retained only those 245 trading days on which all seven stocks and the S&P 500 index had valid prices. This cleaned price panel forms the basis for all subsequent return and risk calculations.

Q2: Risk-Free Benchmark

For the risk-free benchmark, we use the 4-week U.S. Treasury bill yield, identified by the series code TB4WK in the Federal Reserve Bank of St. Louis FRED database. The series was downloaded in Python using the **pandas_datareaderinterface** to FRED for the same calendar window as the equity data (1 September 2024 to 1 September 2025). The raw TB4WK series was resampled to calendar daily frequency and forward filled so that each trading day in the equity panel has an associated risk-free observation. We then reindexed the daily T-bill series to the cleaned stock-price date index, ensuring a one-to-one alignment between the risk-free series and the equity dataset. Finally, we converted the annualised percentage yield into a daily decimal risk-free rate using a 252-trading-day convention, $r_{f,daily} = \frac{TB4WK}{100 \times 252}$. The resulting risk-free **DataFrame** contains 245 daily observations that match the equity panel exactly in terms of dates and is used to compute excess returns, Sharpe ratios and CAPM betas in subsequent sections.

Q3: Stock Selection Rationale

As per the goal of WBS Investments to develop highly diversified trading books of trading it selected seven major and liquid companies from the technology, finance, and healthcare sectors, along with the S and P 500 benchmark. Ever since the emergence of Apple and Microsoft as the leading global technology experts, the consumer technology, enterprise software, and cloud services sectors have been represented. Besides, NVIDIA was included for its outstanding contribution to semiconductor manufacturing and AI-related computing, while Amazon serves the purpose of retail and cloud infrastructure exposure via AWS. JPMorgan Chase stands for the US banking industry and brings in credit conditions and capital markets activity, Barclays adds the UK's and EU's financial exposure, which in turn helps to diversify the risks stemming from different regulations and economic conditions.

Johnson and Johnson has been identified as the defensive stock in the healthcare sector, as there is a stable demand for its pharmaceuticals and medical devices. In summary, this stock universe is a blend of high-growth tech companies, geographically diversified financial institutions, and a defensive health care sector that together offer portfolio diversification at the cost of concentration in any single sector or area.

Q4: Statistical Profiling and Performance

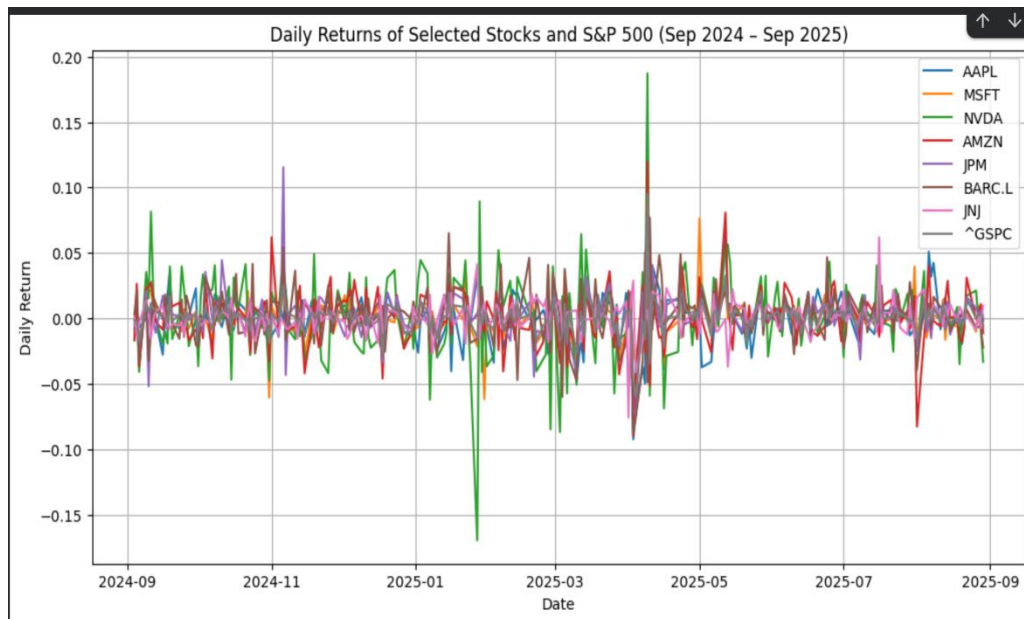
4.1 Descriptive Statistics of Daily Returns

Descriptive Statistics of Daily Returns:						
	AAPL	MSFT	NVDA	AMZN	JPM	BARC.L \
count	244.000000	244.000000	244.000000	244.000000	244.000000	244.000000
mean	0.000390	0.001026	0.002475	0.001299	0.001527	0.002185
std	0.020276	0.015690	0.032051	0.021288	0.017442	0.020772
min	-0.092456	-0.061809	-0.169682	-0.089791	-0.074838	-0.087110
25%	-0.007026	-0.005918	-0.013541	-0.008883	-0.005802	-0.009373
50%	0.001149	0.001127	0.003427	0.000695	0.001644	0.003118
75%	0.009128	0.008092	0.020954	0.013203	0.009749	0.012824
max	0.153289	0.101337	0.187227	0.119770	0.115445	0.076971
	JNJ	^GSPC				
count	244.000000	244.000000				
mean	0.000444	0.000708				
std	0.012250	0.011856				
min	-0.075917	-0.059750				
25%	-0.005523	-0.003088				
50%	0.001050	0.000963				
75%	0.006592	0.005741				
max	0.061932	0.095154				

The yearly table of Descriptive Statistics for Daily Returns verifies the main risk-reward characteristics of the different assets: NVIDIA (NVDA) clearly stands out, showing the most volatility (std: 0.0321) and the highest average daily return (mean: 0.0025), supporting the risk–return trade-off observed in financial markets, while also being affected by the largest single-day movement (max gain of +18.72% and max loss of −16.97%).

On the other hand, the S&P 500 and Johnson & Johnson (JNJ) were the least volatile assets (std: 0.0119 and 0.0123, respectively), thus having the lowest risk profile for investors whose main concern is stability.

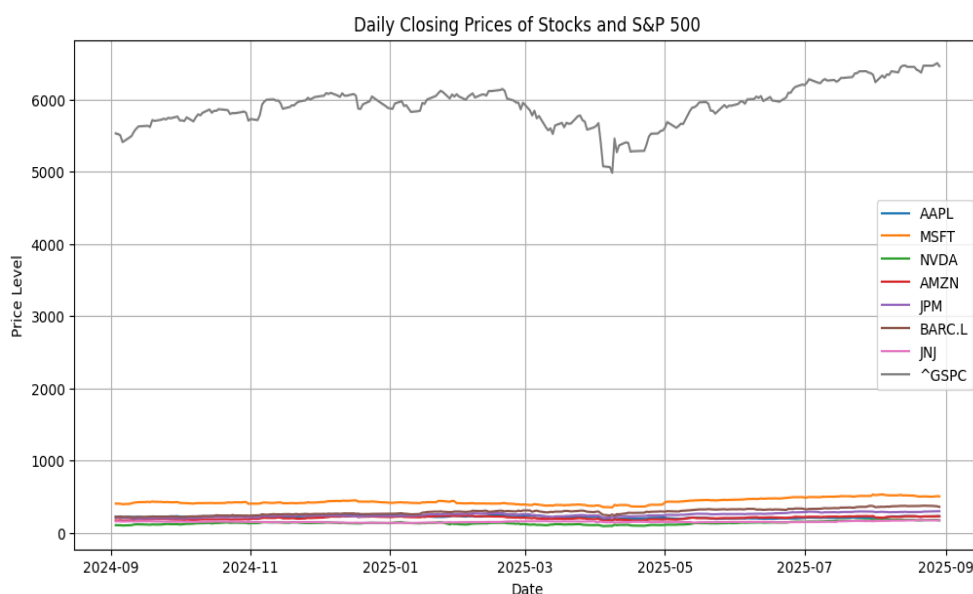
4.2 Daily Returns of the Stocks



The figure illustrates the Daily Returns of the eight major financial assets (Apple, Microsoft, NVIDIA, Amazon, JPMorgan, Barclays, Johnson & Johnson, and the S&P 500 Index) over the period from September 2024 to September 2025.

The chart shows significant volatility, with daily percentage changes mostly ranging between 5% and +5%. In contrast, individual stocks such as NVIDIA (NVDA) and Microsoft (MSFT) experienced extreme daily movements, with gains of up to 15% and losses of approximately 18%.

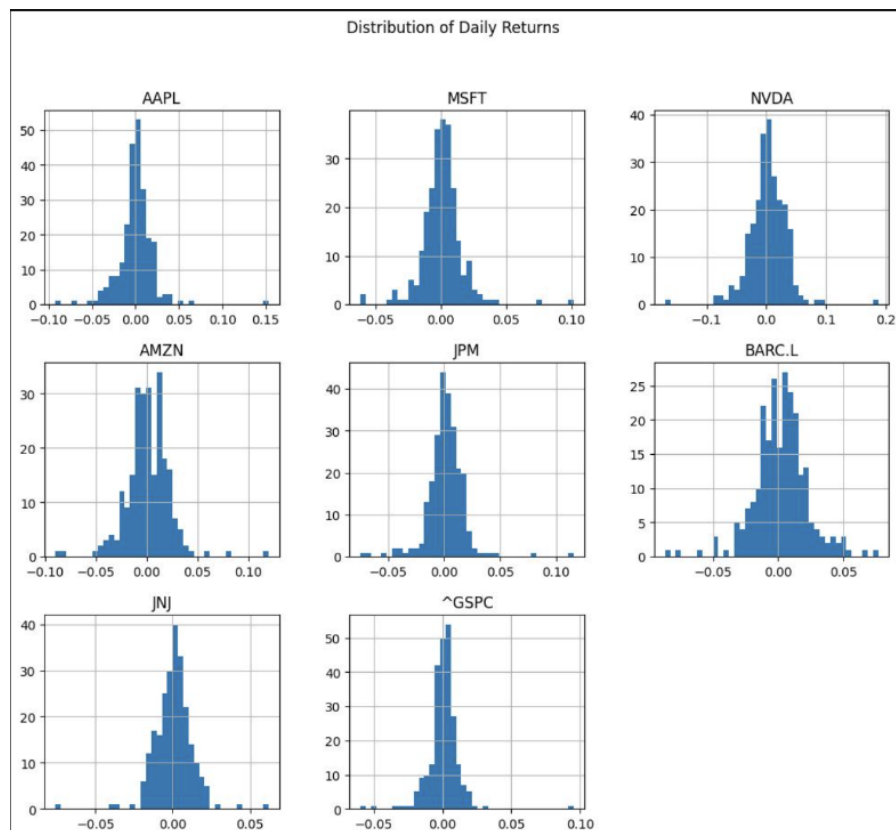
4.3 Daily Closing Prices of the stocks



This diagram shows the Daily Closing Prices for eight important assets: AAPL, MSFT, NVDA, AMZN, JPM, BARC.L, JNJ, and the S&P500 for the period of September 2024 to September 2025, thereby indicating the absolute price change throughout the year.

In the chart, the S&P 500 is placed at the top of the range (between 5,000 and 6,500), showing a strong overall upward trend except for the steep market correction around March/April 2025 that affected the entire market.

The individual stocks are situated closely at the bottom (mostly below the 500 level), where Microsoft (MSFT) and Apple (AAPL) hold the highest price levels in the group, and all stocks generally follow the broad market trend, including the temporary sharp decline and subsequent rebound in the spring.



NVDA delivered the highest Annual Return (62.4%) but had the highest risk ($\beta \approx 2.0$, 37% Max Drawdown). BARC.L was the most efficient (Sharpe 1.67). JNJ was the most resilient (-14.5% drawdown) and defensive ($\beta \approx 0.05$). The overall analysis favours diversified portfolios.

The above figure shows the distribution of daily returns in the form of a histogram

4.4 Sharpe Ratio

Sharpe Ratios:

```
AAPL      0.305439
MSFT      1.037894
NVDA      1.226076
AMZN      0.968350
JPM       1.390034
BARC.L    1.669702
JNJ       0.575221
^GSPC     0.947508
dtype: float64
```

Although NVDA was the most volatile (riskiest), it also gained the highest absolute return. In contrast, BARC.L showed to be the most efficient investment with the highest Sharpe Ratio (1.67), thus giving the best risk-adjusted return. AAPL was less efficient (lowest Sharpe Ratio: 0.31). The S&P 500 (^GSPC) delivered strong returns.

4.5 Betas relative to S&P500

Betas relative to S&P 500:

```
AAPL      1.292529
MSFT      0.967599
NVDA      1.973408
AMZN      1.387306
JPM       1.043206
BARC.L    0.473896
JNJ       0.054083
dtype: float64
```

The Beta analysis reveals that NVDA ($\beta \approx 1.97$) has the highest systematic risk among the stocks analysed, and it is almost twice as sensitive to S&P500 movements. Conversely, JNJ (~ 0.05) is very much uncorrelated with the market, and thus it offers diversification. Sensitivity to the market was high for most of the tech stocks, while the financial sector had stocks like BARC.L ($\beta \approx 0.47$) and JPM ($\beta \approx 1.04$) that are linked to the market to different extents.

4.6 Treynor Ratio

Treynor Ratios:

```
AAPL      0.076062
MSFT      0.267168
NVDA      0.316111
AMZN      0.235881
JPM       0.368941
BARC.L    1.161829
JNJ       2.068346
dtype: float64
```

The Treynor Ratio analysis confirms JNJ as the most efficient stock because of its extremely low Beta, offering the highest return per unit of systematic risk. BARC.L also performed well. On the other hand, high-beta stocks like NVDA and AAPL show poorer risk-adjusted returns when considering only market sensitivity.

4.7 Maximum Drawdowns

Maximum Drawdowns:

```
AAPL      -0.331489
MSFT      -0.218257
NVDA      -0.368810
AMZN      -0.294968
JPM       -0.244239
BARC.L    -0.223739
JNJ       -0.144463
^GSPC     -0.189022
dtype: float64
```

As per the Maximum Drawdown analysis, NVDA experienced the largest loss among the peaks and troughs, which was -36.88% , thus portraying its risk of extreme capital loss. On the other hand, the defensive stock JNJ showed the least sensitivity to the negative market influences with the smallest drawdown of only -14.45% .

The maximum loss of all major tech stocks was much more than the S&P 500's largest loss, which was only -18.90% .

4.8 Summary Table

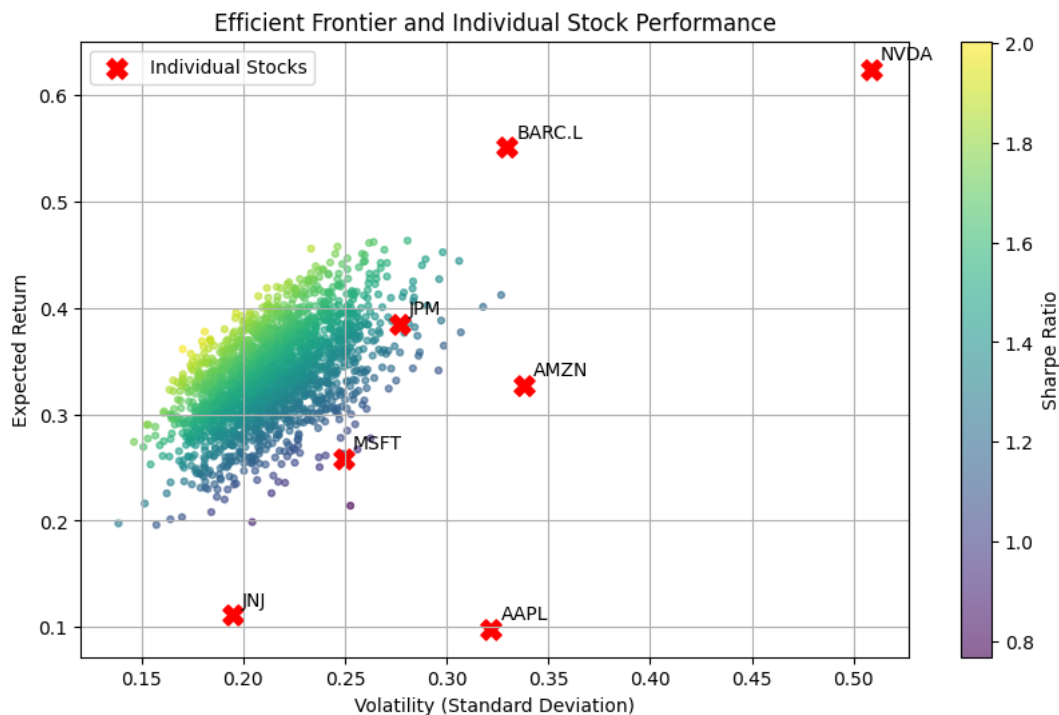
	Annual Return	Annual Volatility	Sharpe Ratio	Max Drawdown
AAPL	0.098312	0.321872	0.305439	-0.331489
MSFT	0.258511	0.249073	1.037894	-0.218257
NVDA	0.623816	0.508791	1.226076	-0.368810
AMZN	0.327240	0.337935	0.968350	-0.294968
JPM	0.384882	0.276887	1.390034	-0.244239
BARC.L	0.550586	0.329751	1.669702	-0.223739
JNJ	0.111862	0.194468	0.575221	-0.144463
^GSPC	0.178334	0.188214	0.947508	-0.189022

The annualised data confirms NVDA delivered the highest Annual Return but also carried the highest Volatility and Max Drawdown. BARC.L was the most efficient on a Sharpe basis, while JNJ was the least volatile asset.

Q5: Portfolio Simulation

Portfolios generated: 2000

To evaluate diversification gains, 2,000 random portfolios were created by picking random proportions for the seven chosen stocks and making sure that portfolio weights added up to one. Anticipated return, risk, and Sharpe ratio were then computed for each portfolio.



The risk-return scatter diagram shows that normally diversified portfolios can command higher returns than the single stocks at the same risk level. Portfolios are represented by different colours according to their Sharpe ratios, wherein warmer shades indicate better risk-adjusted performance. The findings reveal that portfolios with proportions spread evenly over technology, finance, and protective stocks are likely to yield the most efficient results.

The stocks with the highest growth potential, like NVIDIA and Amazon, have been predicted to yield high returns and, at the same time, higher volatility, but would nonetheless be less risky if placed in a mix of portfolios along with the lower-risk assets.

The implications of these results are that diversification not only reduces unsystematic risk but also enables the creation of portfolios that possess better risk-return profiles.

The simulation results will be particularly helpful for the Trading Team in determining portfolio allocations according to different risk preferences.

Q6: Efficient Frontier Analysis

Using simulated portfolios plotted by their expected return and volatility, the efficient frontier was constructed. It displays the portfolios that yield the maximum return for a certain amount of risk, thus being the standard for optimal portfolio selection.

The findings reveal that all single assets are found to be below the efficient frontier, which suggests that investing in single stocks is less profitable than investing in diversified portfolios. Among them, NVIDIA and Apple are placed close to the frontier because they provide high expected returns along with high volatility.

The portfolios that are on the efficient frontier reap the benefits of the ultimate diversification by mixing up assets with different risk–return profiles, thereby enhancing the performance of the entire portfolio. These portfolios get the highest Sharpe ratios, indicating that their risk-adjusted returns are better and they are less exposed to market fluctuations.

In summary, the conclusion is drawn that the Trading Team should give more importance to diversified portfolio strategies over individual stock holdings and should rather beef up their allocations towards the efficient frontier to make their long-term portfolios more efficient and stable.

Q7. Equal-Weight Portfolio:

7.1 Construction of the Equal-Weight Portfolio

Figure 7.1: Key statistics of the equally weighted portfolio.

```
[13] ▶ n = len(stocks)
    equal_weights = np.array([1/n] * n)

    equal_portfolio_returns = stock_returns @ equal_weights

    mean_eq = equal_portfolio_returns.mean()
    var_eq = equal_portfolio_returns.var()
    std_eq = equal_portfolio_returns.std()

    print("Equal-Weight Portfolio Statistics")
    print("Mean Return:", mean_eq)
    print("Variance:", var_eq)
    print("Standard Deviation:", std_eq)

▼ ... Equal-Weight Portfolio Statistics
    Mean Return: 0.0011364517758804416
    Variance: 0.00019723479988951927
    Standard Deviation: 0.014044030756500047
```

To remove personal judgment in portfolio construction and establish a baseline for optimal comparisons, an equally weighted portfolio was formed by allocating equal proportions of investment (1/7 or approximately 14.29%) to each of the stocks short-listed: Apple, Microsoft, NVIDIA, Amazon, JPMorgan Chase, Barclays, and Johnson & Johnson. With equal weights, based on daily log returns, the portfolio's return and standard deviation can be evaluated for performance. As can be noted from Figure 7.1 above, an equal-weighted portfolio provides a balanced performance in terms of volatility and returns.

7.2 Covariance and Correlation Analysis

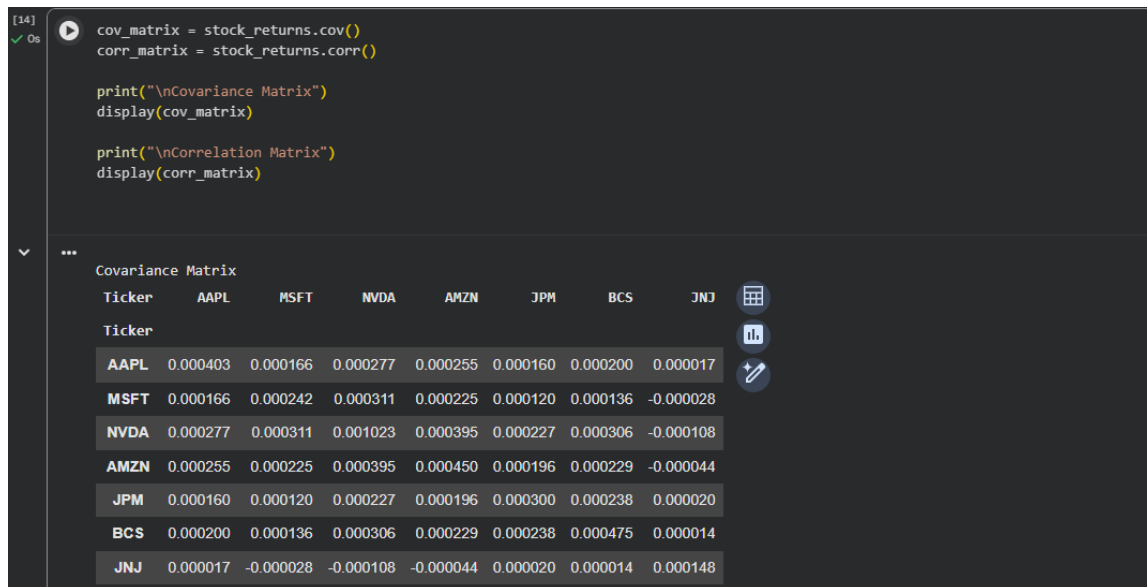


Figure 7.2: Covariance matrix of daily stock returns.

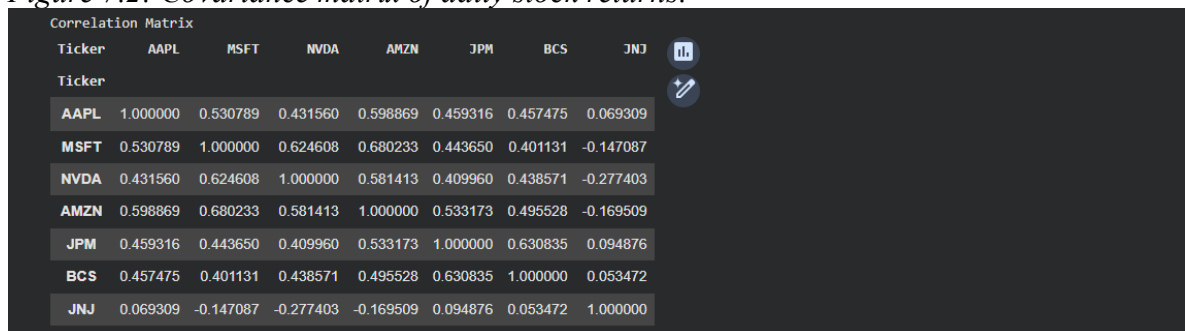


Figure 7.3: Correlation matrix of daily stock returns.

The covariance and correlation matrices are instrumental in understanding linkages among portfolio stocks. Figure 7.2 shows that technology stocks like Apple, Microsoft, NVIDIA, and Amazon have a higher covariance among them, which shows an evident linkage among these stocks. Additionally, Figure 7.2 shows lower covariance among Johnson & Johnson and Barclays stocks and technology stocks, hence emphasising their role in diversification. Moreover, Figure 7.3 depicts a correlation matrix where correlations among stocks in different sectors are lower, thus minimising portfolio risk.

7.3 Efficiency Assessment



Figure 7.3: Equal-weight portfolio compared with simulated portfolios.

In contrast to 2,000 other random portfolio performances, it can be observed in Figure 7.4 that the equal-weighted portfolio plot is inside the feasible risk-return space but not on the efficient frontier. Nevertheless, the competitive analysis above shows how a simple investment approach can attain a decent level of diversification and performance without necessarily using a simulation.

Q8. Optimal Portfolios

8.1 Global Minimum Variance and Maximum Sharpe Portfolios

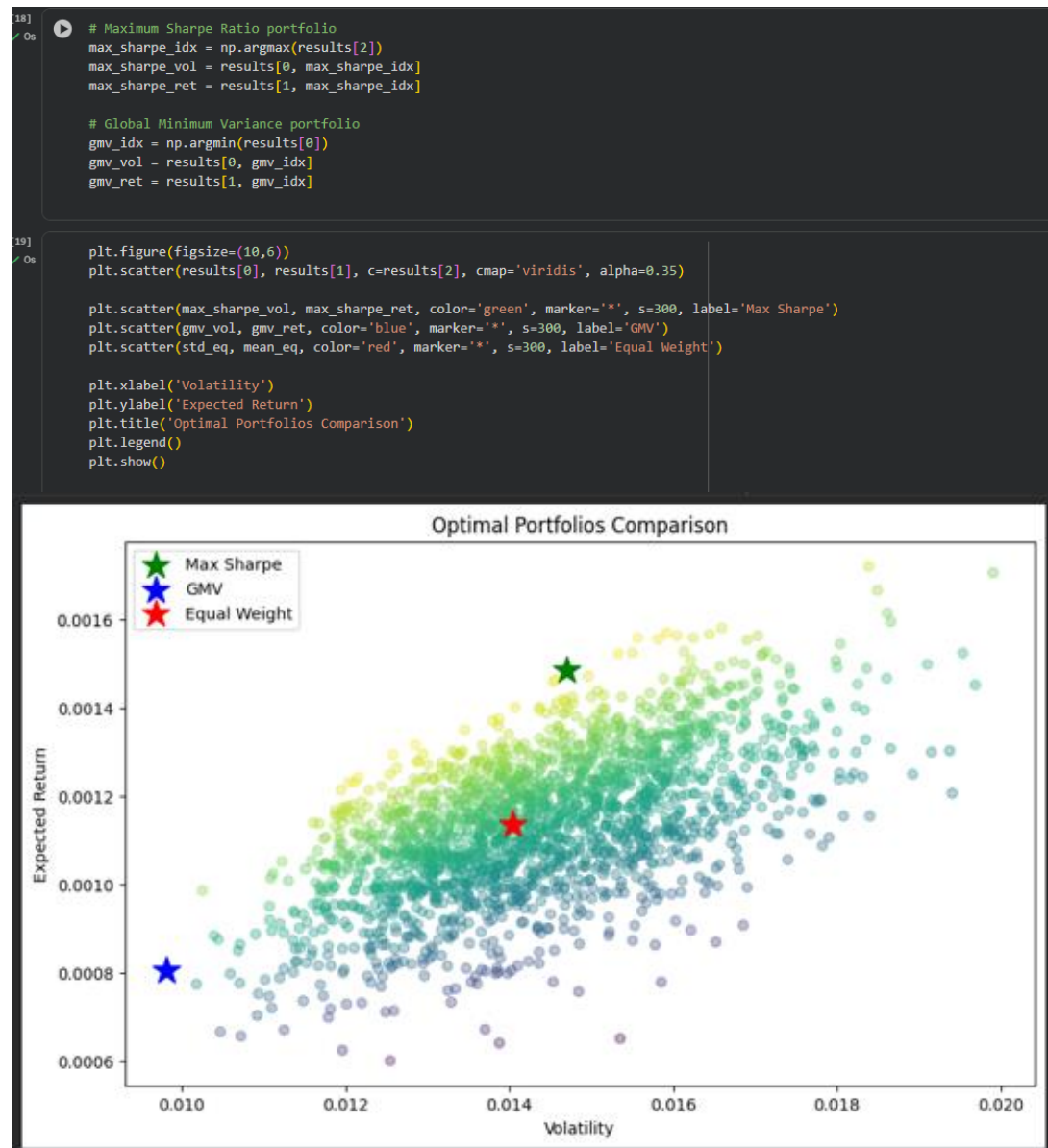


Figure 8.1: Comparison of GMV, Maximum Sharpe, and Equal-weight portfolios.

To illustrate an efficient portfolio composition, a Global Minimum Variance portfolio and a Maximum Sharpe Ratio portfolio were obtained from an analysis of simulated portfolios. While a Global Minimum Variance portfolio is concerned with minimising portfolio risk, with a focus on increasing portfolio weights for less volatile assets with lower correlations, this portfolio will provide a rather low return against portfolio risk as portrayed in Figure 8.1 below. On the other hand, the Maximum Sharpe Ratio portfolio is optimised for maximum risk/return performance. Such a portfolio gives more weight to high-return technology stocks such as NVIDIA, Apple, and Microsoft, resulting in a more volatile portfolio but with higher returns. Therefore, this portfolio will lie at the point of tangency on the efficient frontier among all other simulated portfolios.

When contrasting these optimised portfolios with respect to an equal-weight portfolio, a divergent investment strategy can be observed. While an equal-weight portfolio is suitable for diversification and simplistic investment strategies, a GMV portfolio is ideal for risk-averse investment strategies, and a Maximum Sharpe portfolio is ideal for trading strategies based on returns. For WBS Investments, a hybrid investment strategy could prove efficient: a GMV portfolio for preservation mandates and a Maximum Sharpe portfolio for aggressive strategies.

Q9. Capital Market Line

9.1 Capital Market Line and Efficient Frontier



Figure 9.1: Capital Market Line plotted against the efficient frontier.

The Capital Market Line (CML) was calculated based on the average daily risk-free rate obtained from the 4-week US Treasury Bill and the Maximum Sharpe Ratio portfolio, which serves as the market portfolio or tangency portfolio. As indicated in Figure 9.1 below, the Capital Market Line depicts all possible optimised portfolio combinations of the risk-free asset and the market portfolio with the highest return given a given level of risk. The point where the CML is tangent to the efficient frontier will represent the Maximum Sharpe ratio portfolio. A portfolio below the CML will be referred to as an inefficient portfolio since it is possible to achieve a higher return for a given level of risk with a different allocation of assets among the risk-free asset and the tangency portfolio.

In an investment setting, the CML can be used as a tool for allocating capital. Traders can manage portfolio risk by taking leverage or unleveraging positions in the market portfolio based on their risk preferences and market information. In WBS Investments, the CML can be used to promote sound investment behaviour by incorporating risk-free indices with optimised investment in risky assets.

Q10: Security Market Line Analysis

Using daily stock data and the S&P 500 as a market proxy, CAPM betas were calculated via OLS regression on excess stock and market returns. The risk-free rate used was the four-week Treasury bill from FRED, converted to daily to match return frequency.

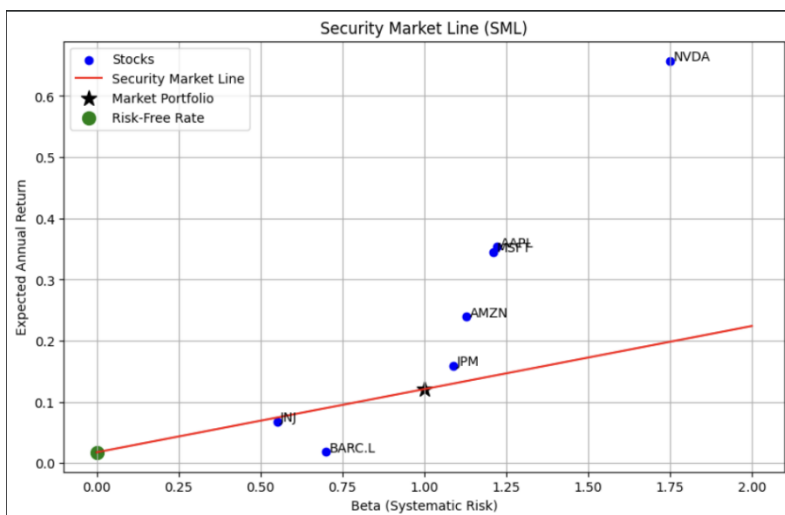
According to the Capital Asset Pricing Model, the systematic risk is associated with the expected return as follows:

$$E(R_i) = R_f + \beta_i(E(R_m) - R_f)$$

where $E(R_i)$ is the expected return on assets i , R_f is the risk-free rate, $E(R_m)$ is the expected market return, and β_i measures the sensitivity of the asset to market movements.

This relationship was used to build the Security Market Line (SML). Stocks that were above the SML, since they yielded higher realised returns than those implied by CAPM, were interpreted as being underpriced; stocks that were below the SML were regarded as overpriced.

It was found that NVDA had the highest annual return and was also the most volatile stock with the highest beta, which reflected both strong performance and high exposure to market fluctuations. AAPL and MSFT also achieved above-average returns, and their betas were above 1. On the opposite side of the scale, BARC.L was the weakest performer and was situated below the SML, which signals that the stock was not performing well in comparison to its risk.



Estimated Betas and Annualized Returns:			
	Stock	Beta	Return
0	AAPL	1.221826	0.353501
1	MSFT	1.211061	0.345246
2	NVDA	1.749493	0.657454
3	AMZN	1.128220	0.238972
4	JPM	1.088221	0.159515
5	BARC.L	0.701042	0.018346
6	JNJ	0.551014	0.068045

Q11 Correlation and Diversification

Correlation indicates the degree to which the prices of different assets move together or apart. If the asset returns are not correlated (or the correlation is negative), the returns of the two assets will be in opposite directions, thus the poor performance of one will be on the other side, the good performance of the other.

Investors practice the concept of diversification by pairing up the assets that are not correlated or negatively correlated, thus reducing the risk of the entire portfolio.

So, for WBS Investments, when they mix their portfolios with the assets that are not very closely related, they will be able to lessen the impact of one asset's price change on their total returns, thus getting a better risk-adjusted return and making the portfolio more stable against the market's ups and downs.

This method is very important in managing risk because it could reduce the chances of incurring large losses in the portfolio during the bearish periods in the market.

Q12: Return and Volatility Forecasting

For studying the short-term market behaviour, first, the strongest and the weakest stocks were named, then selected for the analysis of the forecasting. The best performing stock was NVDA, while the weakest performer was BARC.L. Then, daily returns were computed and divided into training and testing samples.

Through the Box-Jenkins methodology, ARIMA models were implemented to make a forecast on the short-horizon returns. The one-day and one-week ARIMA forecasts produced small and unstable values, which indicated the limitation of return predictability in the short run. This finding agrees with the noisy quality of the equity return series and implies that one should be careful when interpreting ARIMA forecasts.

To obtain the time-varying risk, a GARCH (1,1) model was introduced to the return series. The GARCH model's equation for conditional variance is as follows:

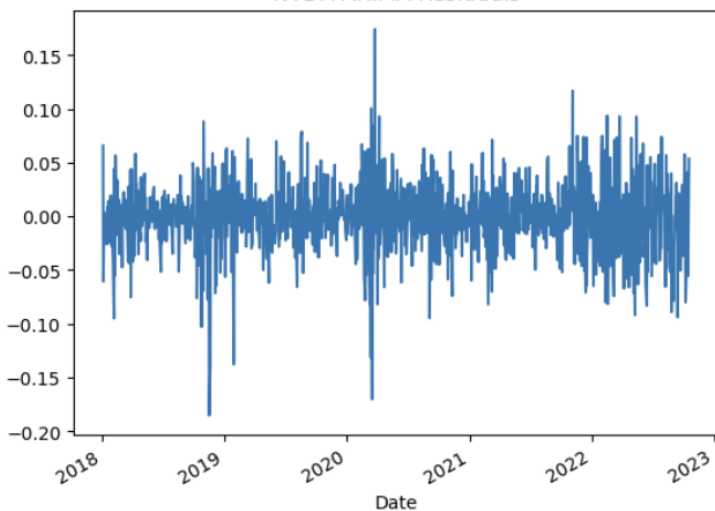
$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

where σ_t^2 represents conditional variance, ε_{t-1}^2 captures the impact of past shocks, and σ_{t-1}^2 represents volatility persistence.

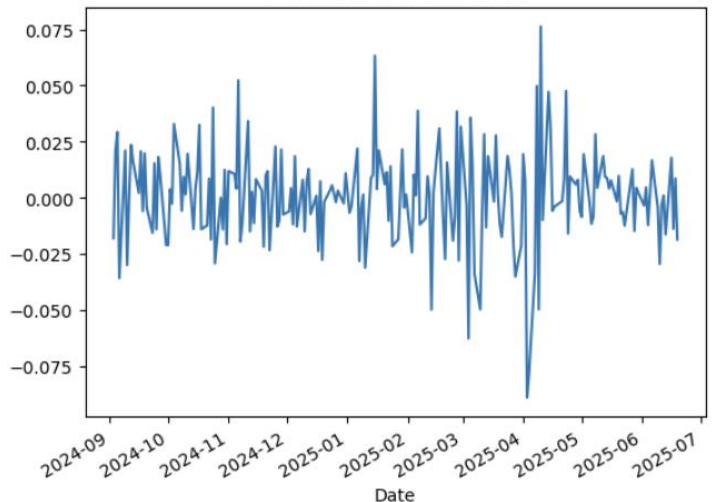
The graphs of the conditional volatility revealed the clear presence of volatility clustering. The NVDA's were more volatile, and that volatility lasted longer than that of BARC.L, which mirrored the latter's risk appetite. BARC.L, on the other hand, had very little and stable volatility, but that came at the cost of being weak in the case.

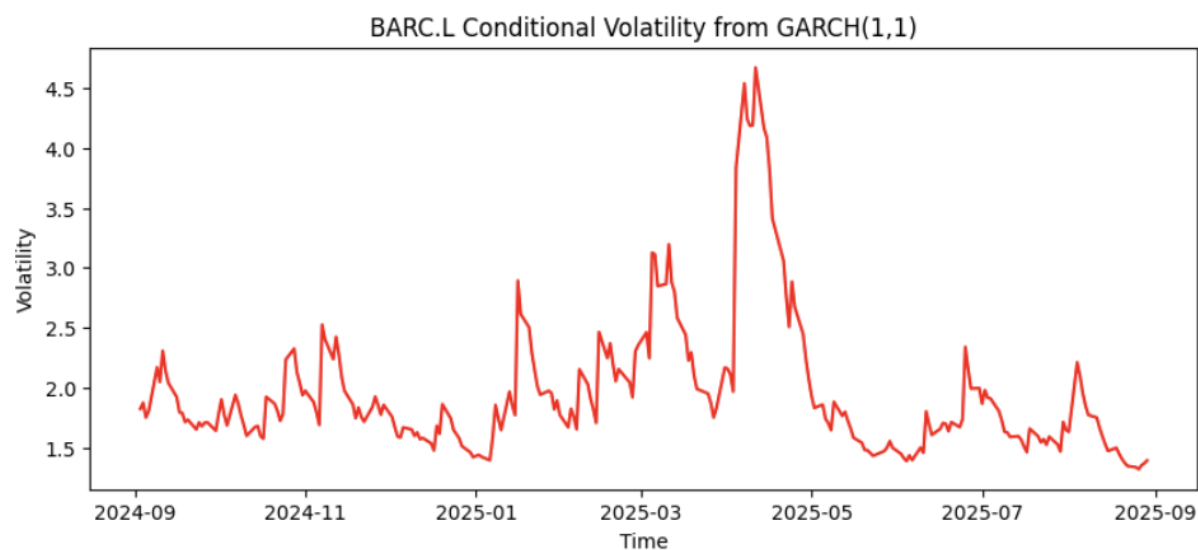
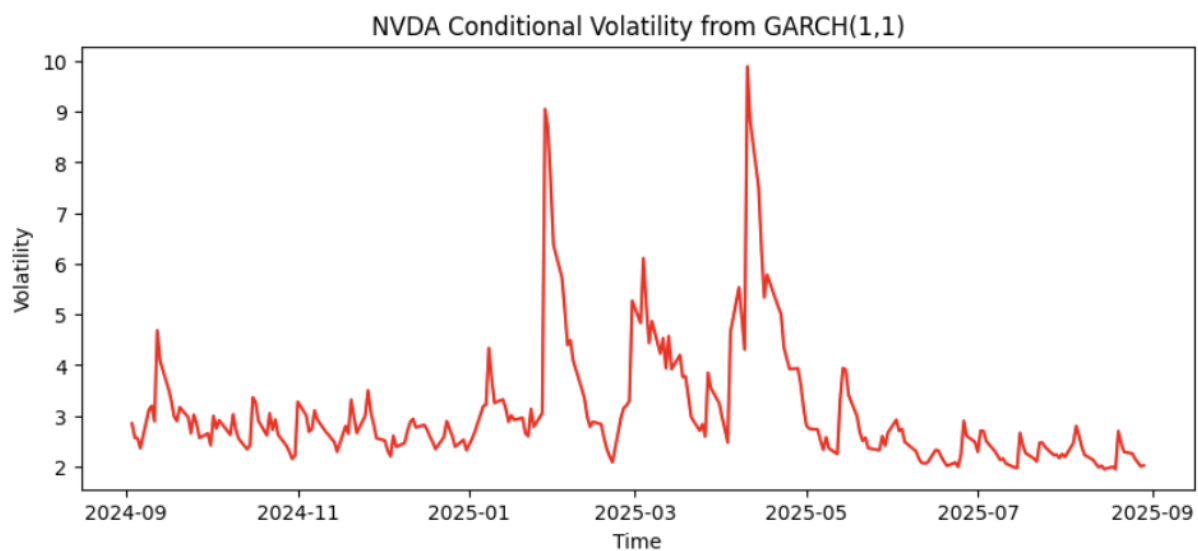
The residuals are plotted over the respective training samples used for each ARIMA model. Since the number of observations differs across stocks, the time ranges of the residual plots differ accordingly

NVDA ARIMA Residuals



BARC.L ARIMA Residuals





Q13: Strategic Recommendations and Reflection

The results show that to make the right decisions regarding a portfolio, the investors should consider the return potential side by side with the risk. The SML analysis emphasises the need of having diversified investments in assets with a wide range of beta values and different sector exposures. For instance, if the investor mixes high-beta growth stocks like NVDA with low-beta stocks like JNJ or BARC.L, the portfolio risk can be mitigated to some extent.

The prediction results point to the fact that the ability to forecast returns in the short term is limited, while the volatility predictions provide more applicable data to risk management. The larger volatility in NVDA suggests higher trading possibilities but also a greater downside risk; on the other hand, the smaller volatility of BARC.L may attract more conservative investors.

In conclusion, CAPM, ARIMA, and GARCH models give important insights and, at the same time, impose limitations. Hence, these quantitative techniques should be regarded as supportive tools along with market judgment, not as independent predictors.

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Appendix

Question 1,2,3

```
!pip install yfinance
```



```

import yfinance as yf
import pandas as pd

tickers = ["AAPL", "MSFT", "NVDA", "AMZN", "JPM", "BARC.L", "JNJ", "AGSPC"]

data = yf.download(
    tickers=tickers,
    start="2024-09-01",
    end="2025-09-01",
    auto_adjust=True,      # adjust for splits/dividends
    progress=False         # no progress bar
)

print(data.head())
print(data.columns)

if isinstance(data.columns, pd.MultiIndex):
    prices = data["Close"]      # or "Adj Close" if not using auto_adjust=True
else:
    prices = data              # fallback if single level

print(prices.head())

# === BASIC INFO ===
print("Prices shape (rows, assets):", prices.shape)
print("Date range:", prices.index.min(), "->", prices.index.max())
print("Tickers:", list(prices.columns))

# Missing values per asset
print("\nMissing values per asset:")
print(prices.isna().sum())

# Drop any dates where at least one asset is missing
prices_complete = prices.dropna(how="any")

print("\nOriginal number of rows:", len(prices))
print("Rows with NO NaNs:", len(prices_complete))

if len(prices_complete) == len(prices):
    print("✅ Dataset is fully consistent across all assets (no NaNs).")
else:
    print("❌ Some dates have missing values, e.g. different US/UK holidays.")
    print("   For later analysis we will use only the fully complete dates.")

# If you want to enforce a perfectly consistent dataset for everything else:
prices = prices_complete

print("\nFinal head of cleaned prices:")
print(prices.head())

!pip install pandas_datareader

# STEP 2 - RISK-FREE BENCHMARK (TB4WK FROM FRED)

import pandas_datareader.data as web

# Same dates as before
start_date = "2024-09-01"
end_date = "2025-09-01"

# 1. Download TB4WK from FRED
tbill = web.DataReader(
    "TB4WK",      # 4-Week Treasury Bill Secondary Market Rate, Discount Basis
    "fred",
    start_date,
    end_date
)

print("Raw TB4WK head:")
print(tbill.head())

# 2. Convert to daily frequency (calendar days) and forward-fill
tbill_daily = tbill.resample("D").ffill()

print("\nTB4WK daily (calendar) head:")
print(tbill_daily.head())

# 3. Align to our cleaned equity price dates
rf = tbill_daily.reindex(prices.index).ffill()

# 4. Convert annualised percentage yield to DAILY decimal rate
trading_days_per_year = 252
rf_daily = rf["TB4WK"] / 100.0 / trading_days_per_year

rf_df = pd.DataFrame({
    "TB4WK_annual_percent": rf["TB4WK"],
    "rf_daily_decimal": rf_daily
})

print("\nAligned risk-free series (same index as prices):")
print(rf_df.head())
print("\nShapes - prices vs risk-free:")
print("Prices:", prices.shape)
print("Risk-free:", rf_df.shape)

```